

Additional file 5

Supplementary figures supporting ctMV-GCN analyses

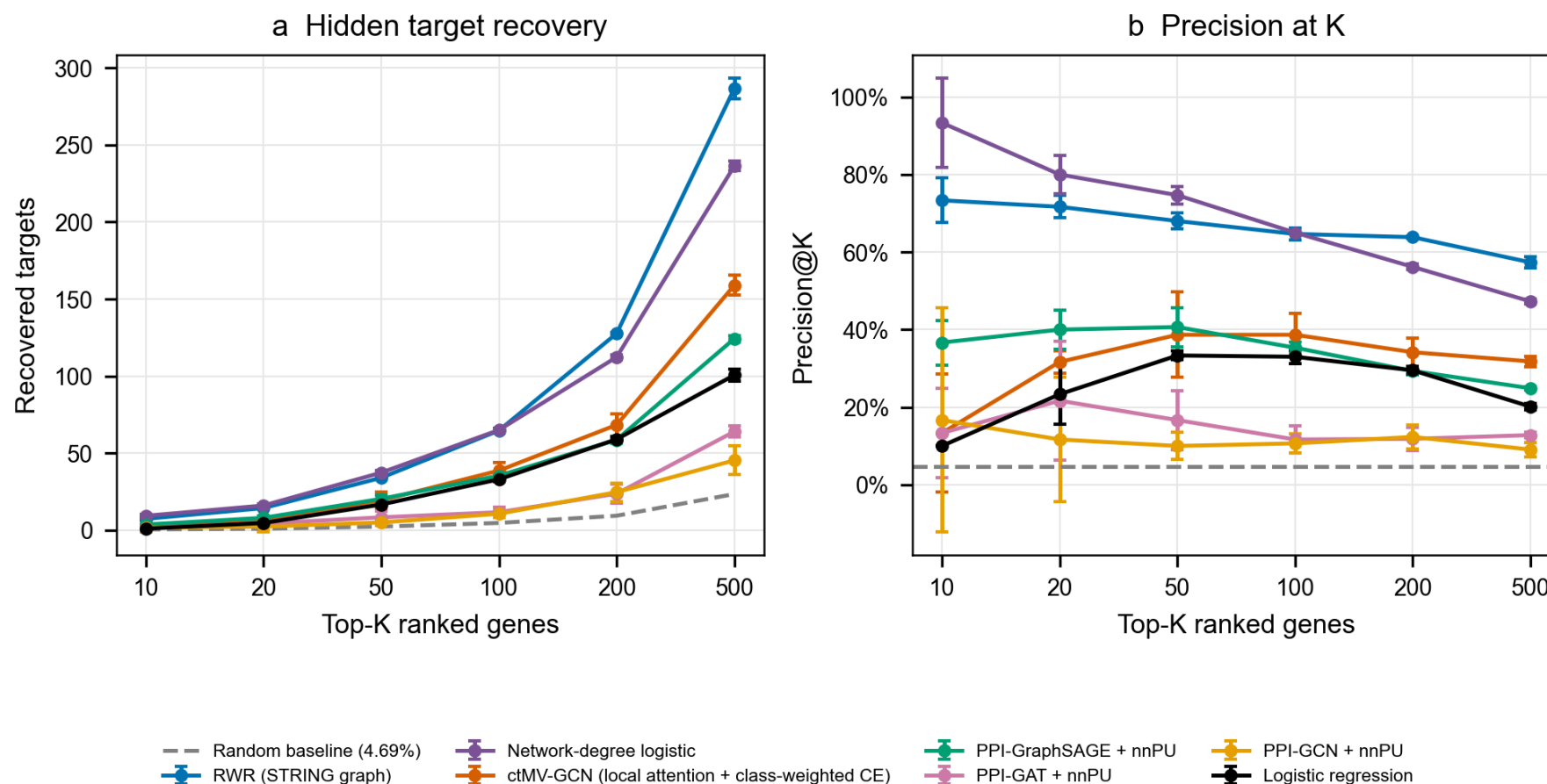
Companion to: ctMV-GCN: cell-type-informed multiview graph convolution for lung adenocarcinoma gene prioritization—a transductive computational study

This file contains six secondary visualizations that support the reported robustness, degree-bias, enrichment-sensitivity and paired-expression analyses. It introduces no new candidate-selection rule, endpoint or numerical result. Complete machine-readable values remain in Additional files 1–4.

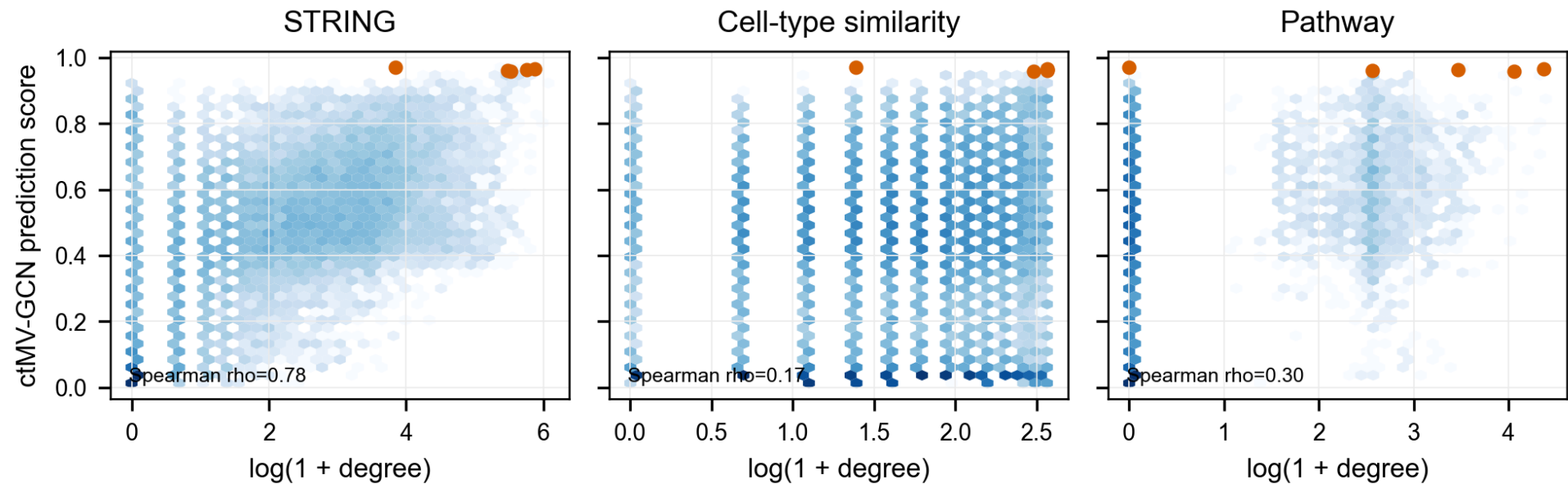
Figure	Content
S1	Top-K recovery in repeated cross-validation
S2	Association between ctMV-GCN scores and network degree
S3	Nonredundant GO-BP enrichment in the primary ranking
S4	GO-BP enrichment after hub and structural-gene filtering
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S6	Paired TCGA expression distributions for four candidates

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Supplementary Figure S1. Top-K recovery in repeated cross-validation

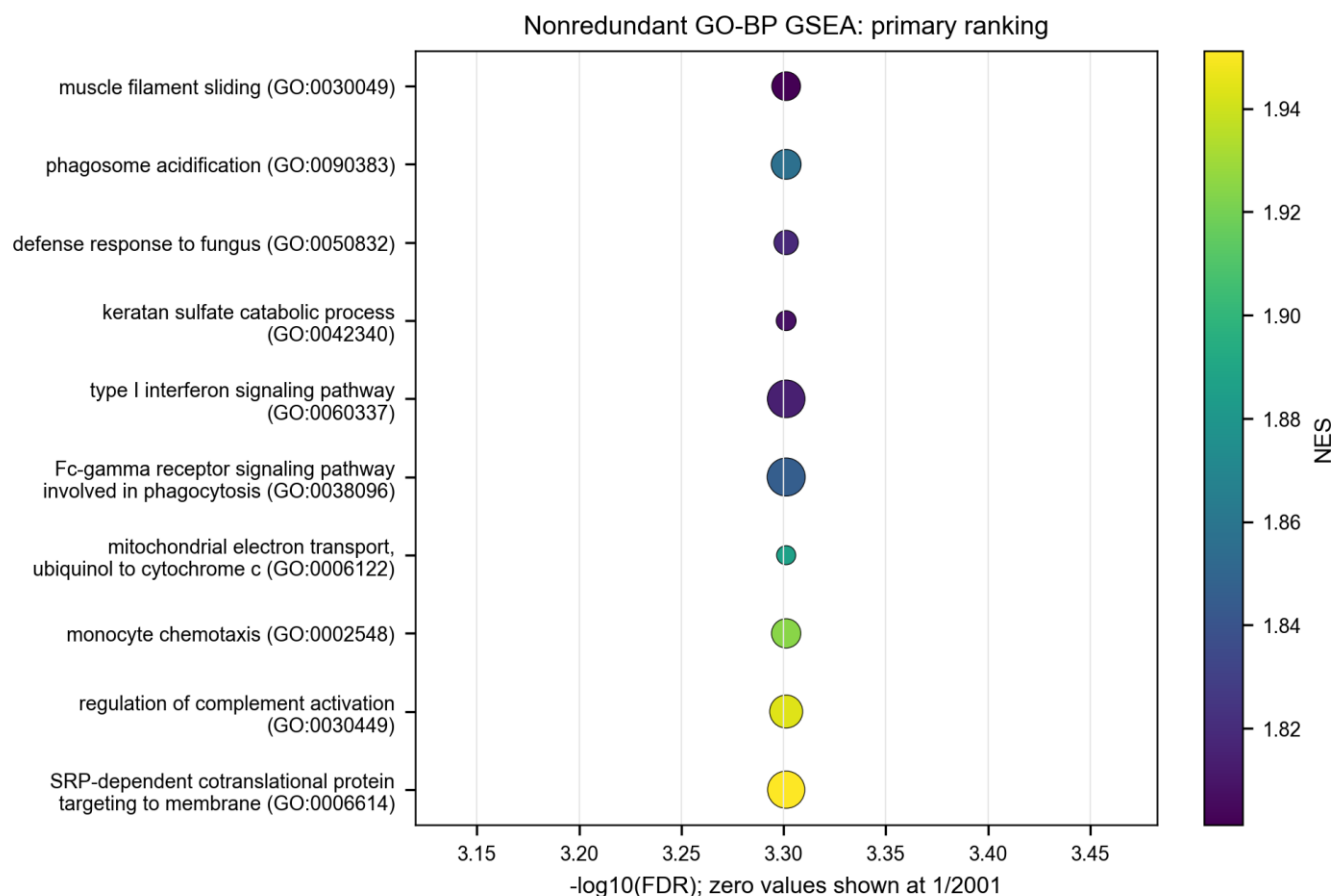


Supplementary Figure S1. Top-K recovery in repeated cross-validation. Out-of-fold predictions were pooled separately within each of the three repeated five-fold outer cross-validation runs. Points show the mean across repeats and error bars show the standard deviation across repeats. (a) Number of hidden positive targets recovered among the top K ranked genes. (b) Precision at K. The dashed baseline uses the observed positive prevalence of 4.69%. Every gene was evaluated only when assigned to an outer test fold; inner-validation labels were not counted as hidden test targets. Complete numerical results are provided in Additional file 1.

Supplementary Figure S2. Association between ctMV-GCN scores and network degree**Network-degree bias assessment**

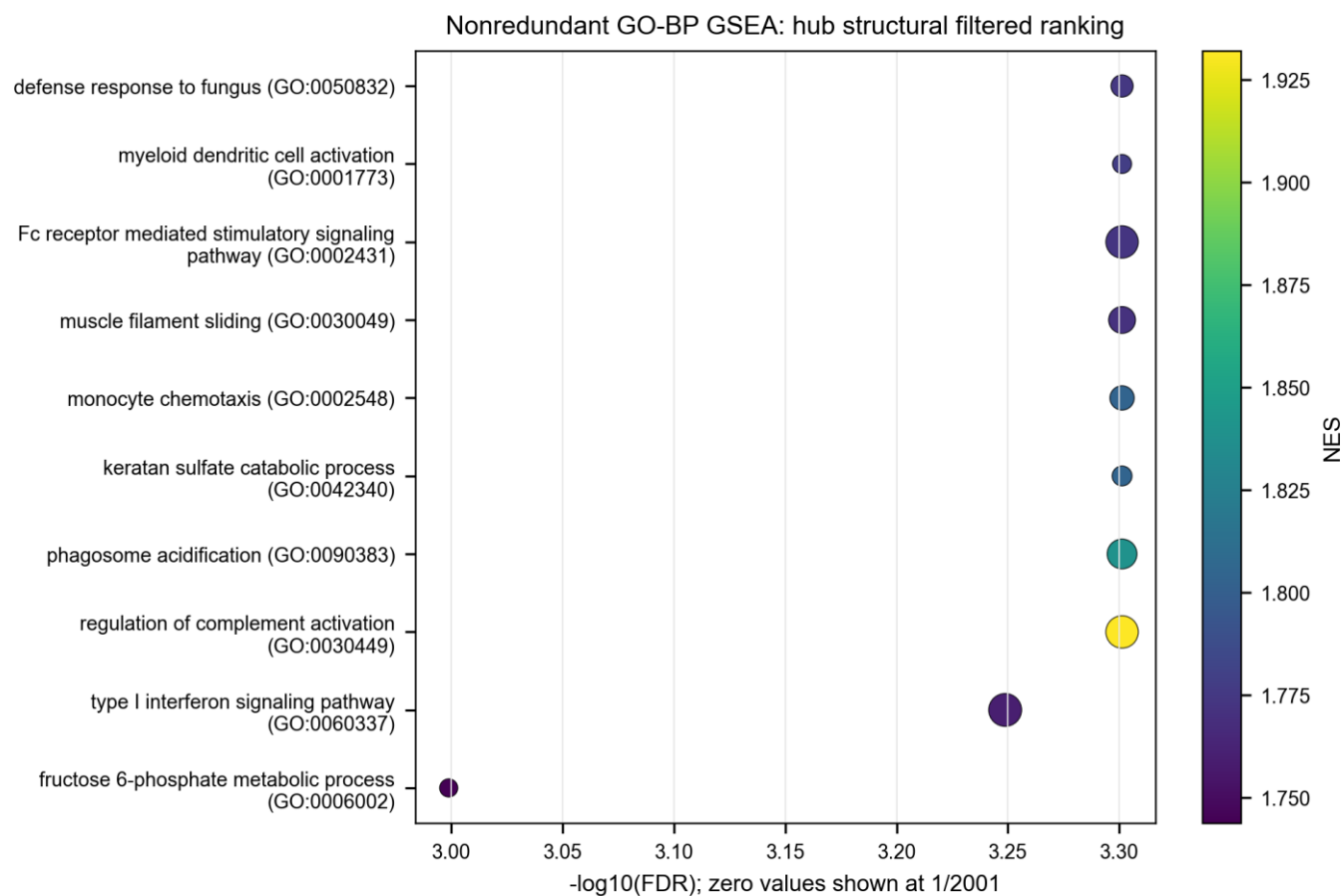
Supplementary Figure S2. Association between ctMV-GCN scores and network degree. Hexagonal bins show the density of unlabeled candidate genes in the STRING functional-association, cell-type similarity and pathway views. The horizontal axis is $\log(1 + \text{unweighted degree})$, and the vertical axis is the mean ctMV-GCN prediction score across ten fixed-seed models. Orange points mark the five highest-ranked genes in the unfiltered mean-score ranking. Each panel reports the Spearman rank correlation. The strong STRING association motivated the degree-matched, conditional-permutation and residualization analyses reported in the main article.

Supplementary Figure S3. Nonredundant GO-BP enrichment in the primary ranking



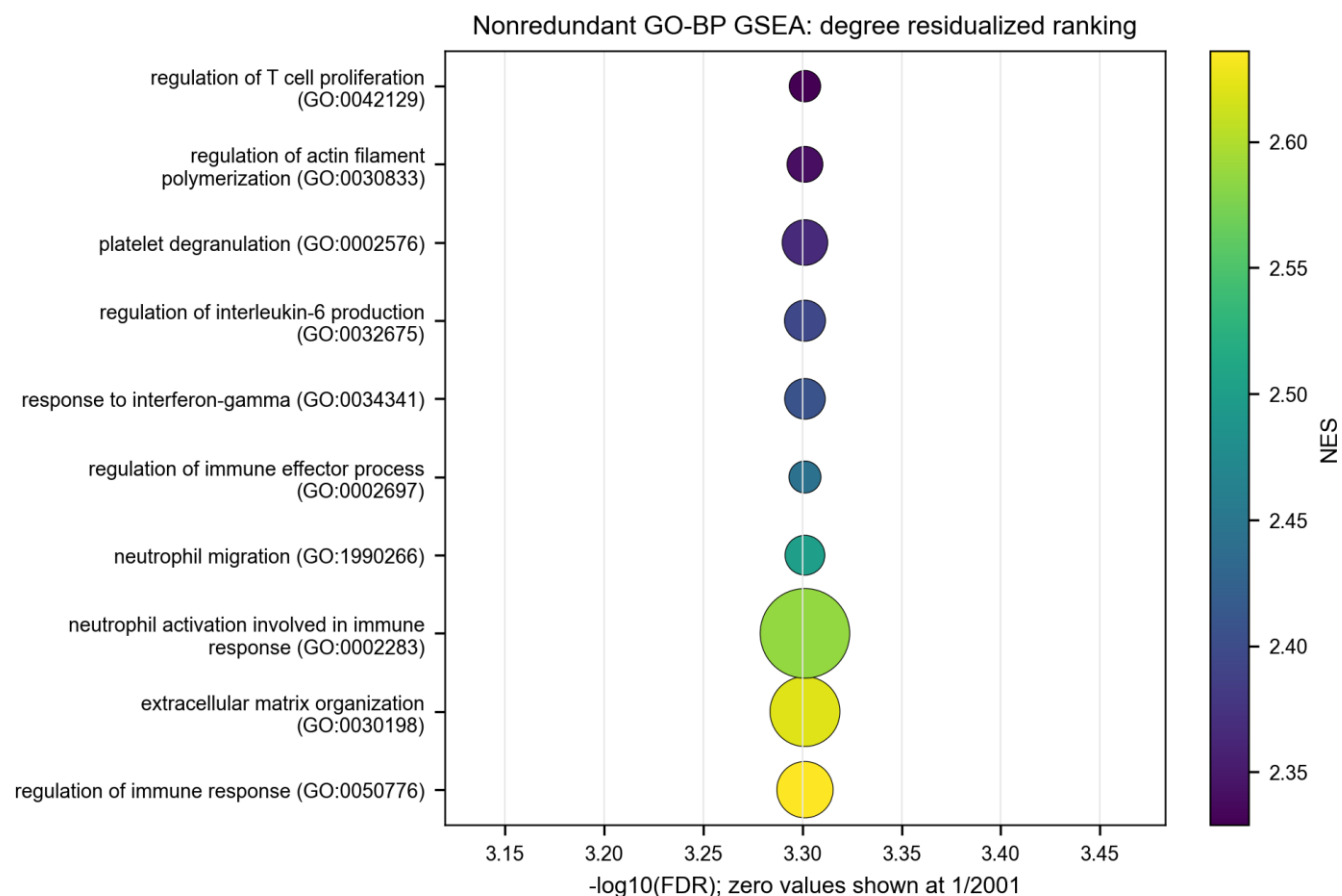
Supplementary Figure S3. Nonredundant GO-BP enrichment in the primary ranking. The plot shows up to ten positively enriched Gene Ontology Biological Process (GO-BP) representatives from preranked GSEA of the primary mean ctMV-GCN score. Significant terms had a normalized enrichment score (NES) > 0 and false-discovery rate (FDR) < 0.05. Representative terms were selected by greedy clustering using a leading-edge overlap coefficient of at least 0.50; the complete uncollapsed result table was retained. Horizontal position is $-\log_{10}(\text{FDR})$, color is NES and bubble area is proportional to the leading-edge hit count. Raw FDR values of zero are displayed using the permutation floor 1/2001.

Supplementary Figure S4. GO-BP enrichment after hub and structural-gene filtering



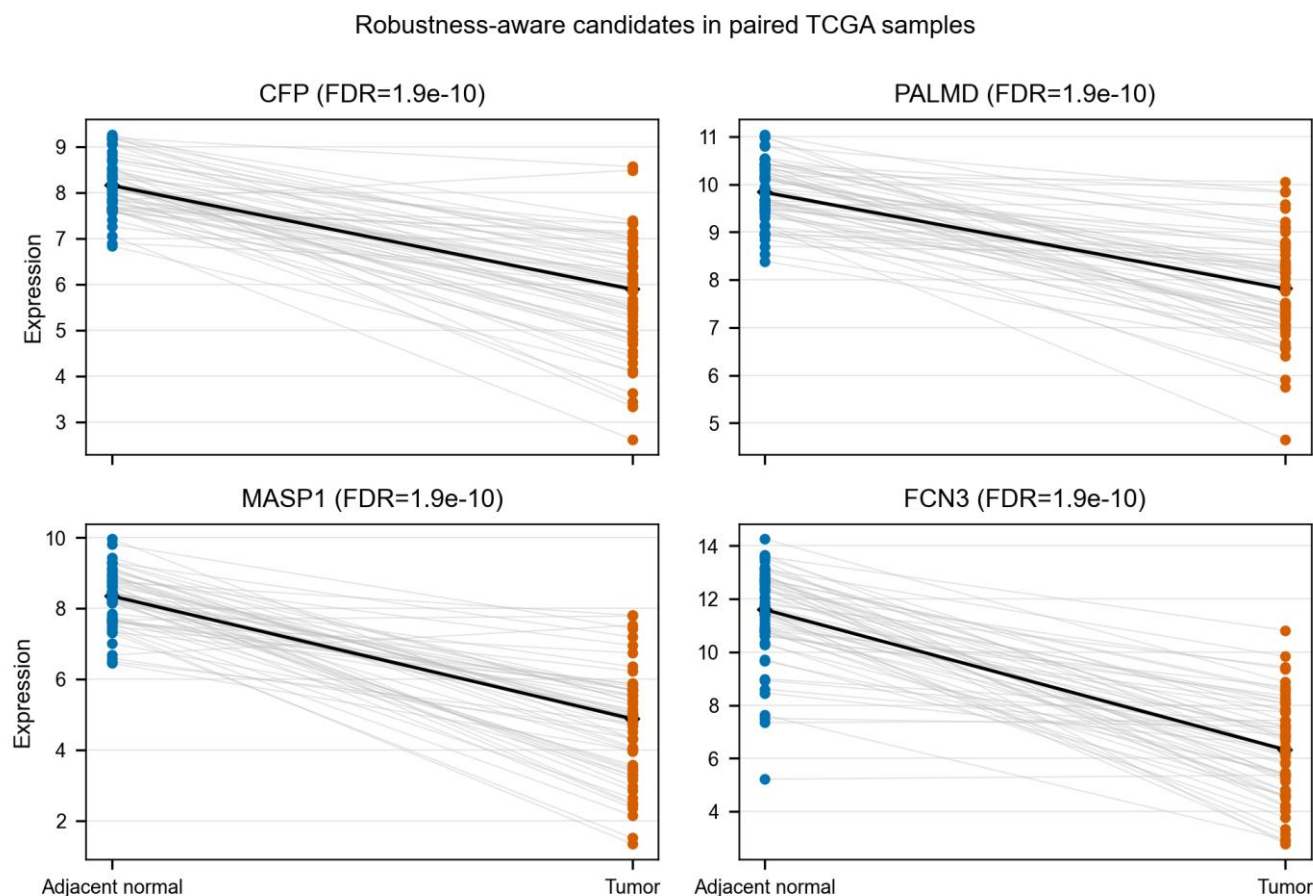
Supplementary Figure S4. GO-BP enrichment after hub and structural-gene filtering. The plot shows up to ten positively enriched nonredundant GO-BP representatives after removing unlabeled genes in the top 5% of STRING degree and genes matching the declared ribosomal or mitochondrial structural-symbol pattern. Significant terms had NES > 0 and FDR < 0.05. Horizontal position is -log10(FDR), color is NES and bubble area is proportional to the leading-edge hit count. Representative-term clustering and the FDR display floor were defined as in Supplementary Figure S3.

Supplementary Figure S5. GO-BP enrichment after nonlinear degree residualization



Supplementary Figure S5. GO-BP enrichment after nonlinear degree residualization. The plot shows up to ten positively enriched nonredundant GO-BP representatives obtained after residualizing candidate scores against six log-transformed unweighted and weighted degree features from the three graph views using the declared nonlinear degree model. Significant terms had NES > 0 and FDR < 0.05. Horizontal position is $-\log_{10}(\text{FDR})$, color is NES and bubble area is proportional to the leading-edge hit count. Representative-term clustering and the FDR display floor were defined as in Supplementary Figure S3. Complete positive and negative enrichment results for all rankings are provided in Additional file 3.

Supplementary Figure S6. Paired TCGA expression distributions for four candidates



Supplementary Figure S6. Paired TCGA expression distributions for four candidates. Paired tumor and adjacent-normal expression values are shown for the four robust candidates with the smallest Benjamini-Hochberg-adjusted Wilcoxon signed-rank P values, with robust-candidate rank used as the tie breaker. Gray lines connect samples from the same patient ($n = 58$), blue points show adjacent normal tissue, orange points show tumor tissue and black diamonds joined by a black line show group medians. Panel titles report the within-family FDR. Expression is shown on the analysis scale; values were transformed as $\log_2(x + 1)$ when the source scale exceeded the prespecified threshold. Per-gene effect sizes, confidence intervals and FDR values are provided in Additional file 4.